

Abdullah Ejaz Janjua

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EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

Bachelor of Science in Computer Science (CGPA: 3.80/4.00)

Topi, Pakistan

Sep. 2023 – June 2027

EXPERIENCE

Undergraduate Researcher

June 2025 - November 2025

Intelligent Machines Lab, Information Technology University

Lahore, Pakistan

- Worked under the guidance of Dr. Waqas Sultani on Domain Generalization, specifically investigating methods to improve DETR-based models' generalizability to unseen target domains
- Improved generalization of the model from 34.1 to 38.7 mAP@50 on the LeukemiaAttri dataset by proposing incorporating Discrete Fourier Transform (DFT) with model features.
- Converted my findings into a comprehensive academic paper.

Research Intern

Jan 2025 – Feb 2025

Intelligent Machines Lab, Information Technology University

Lahore, Pakistan

- Worked with Dr. Waqas Sultani's research group, where I helped develop an Unsupervised Domain Adaptation (UDA) framework for microscopy images with source-target domain transfer.
- Proposed and helped with integration of Monte Carlo Dropout (MCD) for uncertainty-aware pseudo-labeling and confidence image-mixing.
- Assisted in writing, running experiments and designing experiments.

Artificial Intelligence Intern

Sep 2024 – Jan 2025

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

Topi, Pakistan

- Worked under the guidance of Dr. Khurram Khan Jadoon, where I developed and implemented object detection and segmentation algorithms for weapon identification.
- Designed and executed a detection algorithm for Foreign Object Detection (FoD A).
- Created and manually annotated a comprehensive weapon dataset for training high-performance segmentation models.
- Achieved competitive performance compared to industry benchmarks in security applications.

PROJECTS

ResNet Implementation

- Implemented ResNet architecture from scratch using PyTorch, including residual blocks and skip connections.
- Built multiple ResNet variants (ResNet-50, ResNet-101, ResNet-152) with configurable depth and block types.
- Trained and validated the model on image classification datasets to verify implementation correctness.
- Managed to achieve 88.44% accuracy on a simple CardDataset without any hyperparameter tuning.
- **GitHub:** [ResNet Implementation](#)

Vision Transformer (ViT) Implementation

- Implemented Vision Transformer architecture from scratch in PyTorch based on the original paper.
- Built multi-head self-attention mechanism, positional encoding, and patch embedding layers.
- Implemented transformer encoder blocks with layer normalization and feed-forward networks.
- Validated the correctness of the implementation by training on PascalVOC2012, Where I managed to achieve 70.66% accuracy without any pretraining or hyperparameter tuning.
- **GitHub:** [Vision Transformer Implementation](#)

DINOv1 implementation for ResNet

- Implemented the DINOv1 training pipeline for ResNet models, enabling self-supervised feature learning without labels.
- Developed a finetuning pipeline to adapt the pre-trained model for classification tasks.
- Utilized clear coding practices, allowing for seamless training of the model from the command line interface (CLI) without modifying the code.

- **GitHub:** [DINOv1-ResNet](#)

Detection and Segmentation of Weapons with YOLOv8

- Designed a system for detecting and segmenting weapons in images using YOLOv8.
- Worked with a small dataset (100 images) and explored model modifications.
- Performed hyperparameter tuning, including added/removed C2f layers to investigate depth effect.
- Achieved an mAP@50 of 0.82.
- Created segmentation masks using Roboflow.
- Achieved an improved mAP@50 of 0.84.
- **GitHub:** [Weapon Detection and Segmentation](#)

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, HTML, CSS

Frameworks: PyTorch, Flask, Bootstrap

Developer Tools: Git, VS Code, Jupyter Notebook, IntelliJ

Libraries: NumPy, Pandas, Matplotlib, OpenCV, Scikit-learn, Pytorch

COURSEWORK

University Courses: Object-Oriented Programming, Data Structures & Algorithms, Linear Algebra, Statistics, Linear Algebra II, Calculus I

Self-Study: CS50x: Introduction to Computer Science (Prof. David Malan), Machine Learning Specialization (Prof. Andrew Ng), CS231n: Deep Learning for Visual Recognition (Prof. Fei-Fei Li), Computer Vision (Prof. Andreas Geiger)

ACHIEVEMENTS

Dean's Honor Roll: Achieved High Distinction for all semesters so far in recognition of outstanding academic performance. (SGPA ≥ 3.75)

Awarded shield for highest marks obtained in HSSC in my college.